MCP-STYLE INTERFACE FOR THE PUBLIC PROCUREMENT DOCUMENT COMPLETENESS AGENT

Introduction

The Public Procurement Document Completeness Agent assists users in checking procurement documents before submission or evaluation. It identifies whether required information and sections appear to be present and highlights potential omissions. For this task, one existing project capability is **documented as an MCP-style interface**, rather than implemented as a live MCP server.

Selected capability: Procurement Document Completeness Checker. It lets an AI agent or approved application component request a document check and receive a structured result.

Capability Name

check_procurement_document_completeness

Purpose: Analyze an authorized procurement document against a supplied list of required sections. The capability supports review; it does not decide contract awards or automatically reject bidders.

Capability Description

The capability receives a document reference and procurement context, checks the document against approved requirements, and returns found, missing and uncertain sections, a completeness score, a status and recommendations. Its results remain subject to human review.

Inputs

Input	Type	Required	Description
document_id	String	Yes	Unique document identifier
document_name	String	Yes	Uploaded file name
procurement_type	String	Yes	Procurement category
required_sections	Array of strings	Yes	Expected requirements
user_id	String	Yes	Requesting user identifier

Example input

```
{
  "document_id": "PROC-001",
  "document_name": "Supplier_Bid_Document.pdf",
  "procurement_type": "Goods Procurement",
  "required_sections": ["Bidder Information", "Technical Proposal",
    "Financial Proposal", "Tax Clearance Certificate",
    "Declaration Form"],
  "user_id": "USER-102"
}
```

Input Validation

• Confirm the document exists and its format is supported.

• Verify that the requesting user is authenticated and authorized for that document.

• Require a procurement type and a nonempty approved requirements list.

• Reject invalid identifiers, unsupported files and files exceeding the configured size limit.

• Return a clear error without performing an unauthorized check.

Outputs

Output	Type	Description
document_id	String	Document checked
status	String	Overall processing/completeness result
completeness_score	Number	Percentage of requirements confirmed present
found_sections	Array	Requirements identified
missing_sections	Array	Requirements not found
uncertain_sections	Array	Requirements requiring verification
warnings	Array	Additional issues
recommendation	String	Suggested next step

Illustrative output (example data, not a real test result):

```
{
  "document_id": "PROC-001",
  "status": "Incomplete",
  "completeness_score": 80,
  "found_sections": ["Bidder Information", "Technical Proposal",
    "Financial Proposal", "Declaration Form"],
  "missing_sections": ["Tax Clearance Certificate"],
  "uncertain_sections": [],
  "warnings": [],
  "recommendation": "Review and provide the missing certificate."
}
```

Possible Completeness Status Values

• Complete: all required sections were confirmed present.

• Incomplete: one or more requirements were not found.

• Needs Human Review: the result is uncertain and cannot safely be determined.

• Processing Failed: validation, extraction or system processing failed.

Permissions

Authorized roles: approved procurement officers, procurement evaluators, administrators and authorized project users, subject to document-level permission checks.

Allowed: submit a check request, read results, inspect missing-section warnings and detected sections.

Not allowed: delete or modify originals, change approved requirements, access another user's documents, approve or reject bidders, award contracts or transmit documents to unapproved third parties.

Security Boundary

Inside: current authorized procurement document, approved requirement list and minimal temporary processing data.

Outside: unrelated user files, passwords, other users' documents, confidential databases and systems lacking explicit authorization.

The capability applies **least privilege**: only the minimum access necessary for the completeness check is available.

Security Controls

• Authentication before a tool request; authorization on each document.

• File-type, file-size and input validation before processing.

• Read-only handling of original procurement documents.

• Isolation of one user's data from another user's data.

• Treat document text as untrusted data rather than instructions to the agent.

• Audit request, authorization decision and result status without logging unnecessary sensitive content.

Human Approval Boundary

AI findings are advisory. A procurement officer must verify missing or ambiguous items before formal action. Accepting a bid, rejecting a bid, declaring compliance and awarding a contract remain human responsibilities.

Failure Behaviour

The capability returns a structured error and fails safely when validation or processing cannot complete. Example responses:

```
{"status":"error","error_code":"UNSUPPORTED_FILE",
 "message":"The submitted format is not supported."}
{"status":"error","error_code":"ACCESS_DENIED",
 "message":"User is not authorized for this document."}
{"status":"error","error_code":"INVALID_INPUT",
 "message":"required_sections is missing."}
{"status":"Needs Human Review",
 "message":"One or more requirements cannot be verified.",
 "recommendation":"Refer to a procurement officer."}
```

MCP-Style Tool Definition

Tool name: check_procurement_document_completeness

Description: Checks an authorized procurement document against required sections and returns findings in a predictable structured format.

Input schema (illustrative):

```
{
  "document_id": "string", "document_name": "string",
  "procurement_type": "string",
  "required_sections": ["string"], "user_id": "string"
}
```

Output schema (illustrative):

```
{
  "document_id": "string",
  "status": "Complete | Incomplete | Needs Human Review",
  "completeness_score": 0,
  "found_sections": ["string"],
  "missing_sections": ["string"],
  "uncertain_sections": ["string"],
  "warnings": ["string"],
  "recommendation": "string"
}
```

Permission level: authenticated users authorized to read the document. **Security boundary:** read current authorized document and approved requirements; no changes to procurement records or final decisions.

Note: This is an MCP-style interface specification, not a claim that an MCP server or protocol endpoint has been implemented. In a production implementation, authenticated identity should come from the security context rather than trusting a user_id field supplied by the caller.

Example Interaction

• The procurement officer requests a completeness check on an uploaded bid.

• The agent calls check_procurement_document_completeness with the document reference and requirements.

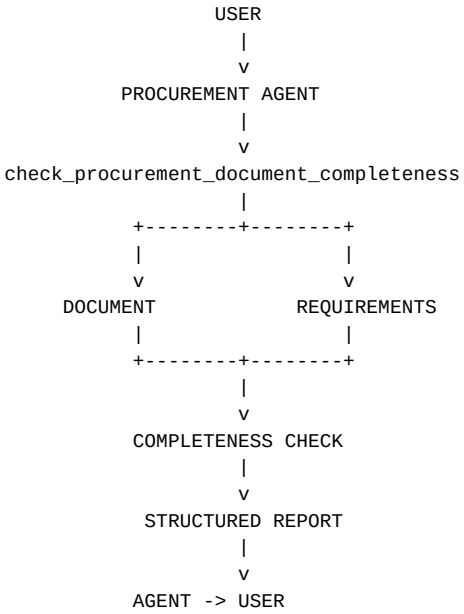
• The capability validates permissions, reads the document and checks each requirement.

• The capability returns four found requirements, one missing requirement and a score of 80% (illustrative example).

• The agent shows the result and requests human verification.

• An authorized officer determines any formal next action.

Simple Architecture



Why This Capability Was Selected

It is the core function of the project, has clear inputs and structured outputs, supports granular permissions and a defined security boundary, and can be invoked by an AI agent while preserving human oversight.

Conclusion

The documented capability defines what the agent may request, what information it must supply, what output it receives and where its authority ends. This satisfies the assignment option to **document a project capability as an MCP-style interface**. It is a design deliverable and should not be presented as a deployed integration unless separately implemented and tested.